III B.TechISemester

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23A20505 ELECTRICAL SAFETY PRACTICES AND STANDARDS
(Open Elective-I)

Course Outcomes:

CO1: Understanding the Fundamentals of Electrical Safety -L2
CO2: Identifying and Applying Safety Components -L3
CO3: Analyzing Grounding Practices and Electrical Bonding
CO4: Applying Safety Practices in Electrical Installations and Environments- L4
CO5: Evaluating Electrical Safety Standards and Regulatory Compliance -L5

UNIT I IntroductionToElectricalSafety:
Fundamentals ofElectricalsafety-ElectricShock-physiologicaleffects of electric current-Safetyrequirements–Hazardsofelectricity-Arc-Blast-Causesforelectricalfailure.

UNIT II Safety Components:
Introduction to conductors and insulators- voltage classification -safety against over voltages- safety againststaticelectricity-Electricalsafetyequipment's-Fireextinguishersforelectricalsafety.

UNIT III Grounding:
General requirements for grounding and bonding- Definitions- System grounding-Equipment grounding -The Earth-Earthingpractices-Determiningsafeapproachdistance-Determiningarchazardcategory.

UNIT IVSafetyPractices:
Generalfirstaid-Safetyinhandlinghandheldelectricalappliancestools-Electricalsafetyintrain stations-swimmingpools,externallightinginstallations,medicallocations-Casestudies.

UNIT V Standards For Electrical Safety:
Electricity Acts- Rules & regulations- Electrical standards-NFPA 70 E-OSHA standards-IEEE standards-National Electrical Code 2005 – National Electric Safety code NESC-Statutory requirements from electricalinspectorate

TEXTBOOKS:

1. MassimoA.G.Mitolo,"ElectricalSafetyofLow-VoltageSystems",McGrawHill,USA,2009.
2. MohamedEl-Sharkawi,"ElectricSafety-PracticeandStandards",CRCPress,USA,2014

REFERENCES:

1. KennethG.Mastrullo, RayA.Jones,"TheElectricalSafetyProgramBook",Jones andBartlettPublishers,London, 2nd Edition, 2011.
2. PalmerHickman,"ElectricalSafety-RelatedWorkPractices",Jones&BartlettPublishers,London,2009.
3. FordhamCooper,W.,"ElectricalSafetyEngineering",ButterworthandCompany,London,1986.
4. JohnCadick,MaryCapelli-Schellpfeffer,DennisK.Neitzel,"ElectricalSafetyHandbook,McGraw-Hill,NewYork,USA, 4th edition, 2012.